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import cv2
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Create a synthetic background (gray image)
bg = np.full((200, 300), 50, dtype=np.uint8)

# Add some static objects to the background
cv2.rectangle(bg, (20, 50), (80, 150), 100, -1)
cv2.rectangle(bg, (220, 30), (280, 120), 120, -1)

# Step 2: Create a frame with a moving object
frame = bg.copy()

# Draw a moving circle (simulating a person/vehicle)
cv2.circle(frame, (150, 100), 20, 255, -1)

# Step 3: Initialize Background Subtractor
backSub = cv2.createBackgroundSubtractorMOG2()
```